1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In breast tumors with high Tumor-Infiltrating Lymphocyte (TIL) density, CXCR4 signaling drives upregulation of GLS (glutaminase) in tumor and/or stroma-associated cells, promoting a glutamine–dependent metabolic state that impairs local T-cell effector function. This metabolic crosstalk creates a microenvironment where T cells are functionally suppressed despite substantial TIL presence, leading to reduced therapy_response to checkpoint inhibitors. Consequently, tumors with the TIL-high and GLS-high phenotype will exhibit worse immunotherapy response than TIL-high/GLS-low tumors. The calculation is that GLS activity (and CXCR4-driven GLS upregulation) can be modulated by secondary modifiers—CCND1 (Cyclin D1), CTNNB1 (β-catenin), and the EMT_signature—which tune CXCR4 expression, glutamine metabolism, and T-cell localization, thereby altering immunotherapy outcomes. This yields a testable model: (i) TIL-high/GLS-high predicts poor therapy_response; (ii) GLS inhibition or CXCR4 blockade will restore T-cell function and improve therapy_response in this subset; (iii) secondary modifiers modulate the strength of this axis and can refine patient stratification for combination therapies.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	GLS, CXCR4
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

GLS: GLS and notes that high GLS can undermine immunotherapy; this raises the question: why do some immune-infiltrated (TIL-high) tumors fail to benefit from immunotherapy? 2) Tier 1 paths used: I built on the EMT→Wnt→CXCR4→GLS axis because it provides a mechanistic route connecting tumor plasticity/chemotaxis (EMT/CXCR4) to metabolic reprogramming (GLS) that can directly affect immune function.

CXCR4: CXCR4 → GLS and notes that high GLS can undermine immunotherapy; this raises the question: why do some immune-infiltrated (TIL-high) tumors fail to benefit from immunotherapy? 2) Tier 1 paths used: I built on the EMT→Wnt→CXCR4→GLS axis because it provides a mechanistic route connecting tumor plasticity/chemotaxis (EMT/CXCR4) to metabolic reprogramming (GLS) that can directly affect immune function.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	CCND1	Supporting modifier from knowledge graph
Secondary	CTNNB1	Supporting modifier from knowledge graph
Secondary	EMT_program	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and EMT programs are associated with immune exclusion and may upregulate CXCR4 and metabolic adaptations, while CCND1 intersects cell cycle with metabolism, potentially sustaining a proliferative, metabolically active niche.

Prediction 2: . If validated, combining GLS inhibitors or CXCR4 antagonists with checkpoint blockade could overcome resistance in the TIL-high/GLS-high subset and refine patient stratification using the proposed anchor genes and secondary modifiers.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CCND1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CTNNB1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.